

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Nuclear Geophysics L-T-P: 3-0-0

Credits:3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Nuclear systematics, naturally occurring radioactive isotopes and series, instrumental methods for detection and measurement of radioactivity, counting statistics, Poisson distribution and error analysis, radioactive methods for prospecting and assaying of mineral (radioactive and non-radioactive) deposits, Air-borne (STOL), Heli-borne, Drone based, Vehicle Borne and Hand-held Radiometric Surveys. Non-invasive Gamma Ray and Radon surveys for shallow and intermediate depths, Radiometric prospecting for beach placers, including titanium, zirconium and rare-earth deposits, recent advances in portable gamma ray spectrometry and radon manometry, modeling of radiometric data, iso-rad mapping and its interpretation, applications of radioactivity and radon in prospecting for oil and hydrocarbon deposits, applications of radiometric studies to paleo seismology, modeling of ground water and coastal geo-environment using radioactive tracers, nuclear fallout and waste disposal, nuclear logging and its applications, basic principles of various radiometric techniques for dating of rocks.

Reference Books:

- 1) Principles of Nuclear Geology by U. Aswathanarayana
- 2) Radioactivity in Geology by E.M. Durance
- 3) The Solid Earth: An Introduction to Global Geophysics by C. M. R Fowler
- 3) Nuclear Radiation Physics by R.E. Lapp and H.L. Andrews
- 4) Applications of Unmanned Aerial Vehicles in Geosciences by Tomasz Niedzielski
- 5) Fundamentals of Capturing and Processing Drone Imagery and Data by Amy E. Frazier, Kunwar K. Singh

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Nuclear Systematics, Rutherford-Soddy's relationship and implications, Single Decay, Branched Decay and Decay Series, Rutherford-Soddy's relationship and its significance	2
2.	Naturally Occurring Nuclear Deposits, Role of	2

	Critical and Strategic Minerals including Rare Earth Deposits, Placer Minerals as economic resources.	
3.	Decay Statistics, Binomial and Poisson Statistics Modelling of Radiometric Data and its interpretation, Significance of Iso-rad Maps and Solid Angle effect, Role of surface topography	2
4.	Radiometric Assaying and Prospecting and its applications, Hand Held, Vehicle borne, Drone based Radiometric Surveys, Recent studies of Drone based data for mineral exploration and reliable inputs for Disaster Management, Heli-borne and Air-Borne Radiometric Surveys, Marine Radiometric surveys and its important applications.	6
5.	Radon Measurements and its applications, Radon as a Seismic precursor, Role of Radon for Indoor/Ambient environment in Dwellings, Industrial Complexes and Radioactivity enriched Mine atmosphere.	3
6.	Radionuclide studies in Groundwater, Ambient Air and in Snow and Ice, Modelling of related data for geological hazard index, contamination/pollution index and parameters like Residence Time, Recharge Rates and detailed ground water modelling	5
7.	Nuclear Logging techniques including NMR, Recent advances in data acquisition and interpretation, Advances in Integrated Geophysics coupled to Non-invasive radiometric data.	5
Total number of hours		25